

A Cooperative Drone Assisted Mobile Access Network for Disaster Emergency Communications

Di Wu*, *Student Member, IEEE*, Xiang Sun†, *Member, IEEE*, and Nirwan Ansari*, *Fellow, IEEE*

*Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ 07102, USA

†Department of Electrical and Computer Engineering, University of New Mexico, Albuquerque, NM 87131, USA

Abstract—Multiple drone-mounted base stations (DBSs) are used to be deployed over a disaster struck area to help mobile users (MUs) communicate with working BSs, which are located beyond the disaster struck area. DBSs are considered as relay nodes between MUs and working BSs. In order to relax the bottleneck in wireless backhaul links, we propose a cooperative drone assisted mobile access network architecture by enabling DBSs (whose backhaul links are congested) to offload their traffic to other DBSs (whose backhaul links are not congested) via DBS-to-DBS communications. We formulate the DBS placement and channel allocation problem in the context of the cooperative drone assisted mobile access network architecture, and design a COoperative DBS plAcement and CHannel allocation (COACH) algorithm to solve the problem. The performance of COACH is demonstrated via extensive simulations.

Index Terms—Drone-mounted base station, emergency communication, self-interference

I. INTRODUCTION

Drones are unmanned aerial vehicles that can be flexibly and quickly deployed over specific areas. Recently, drone-mounted base stations (DBSs) are used to be deployed over disaster struck areas to rapidly recover communications in disaster struck areas [1]–[3]. Here, DBSs are considered as movable small cells acting as relay nodes between working macro base stations (which are located out of disaster struck areas) and mobile users (MUs) in disaster struck areas. There are two types of relay mode for a DBS. 1) Ferry mode: a DBS will fly over a disaster struck area based on a designed itinerary starting from a working macro base station (MBS) and finally back to the same MBS. Meanwhile, the DBS will collect data sent from MUs as it flies over them [4]. 2) Hovering mode: a DBS will hover over a designed location, receive data from MUs via wireless access links, and relay data to a nearby working MBS via the wireless backhaul link. A DBS in the ferry mode normally incur long delay in relaying data from MUs to a working MBS since the delay depends on the time required in completing the itinerary. On the other hand, a DBS in the hovering mode can relay MUs' data in real time [5]–[7]. However, in order to facilitate MUs for transmitting data to DBSs, DBSs may be placed close to MUs, and thus the distance between DBSs and their associated MBSs may be very long. That is, the wireless backhaul link between a DBS and a working MBS may become the bottleneck of uploading

MUs' data to a working MBS, thus significantly reducing the overall throughput.

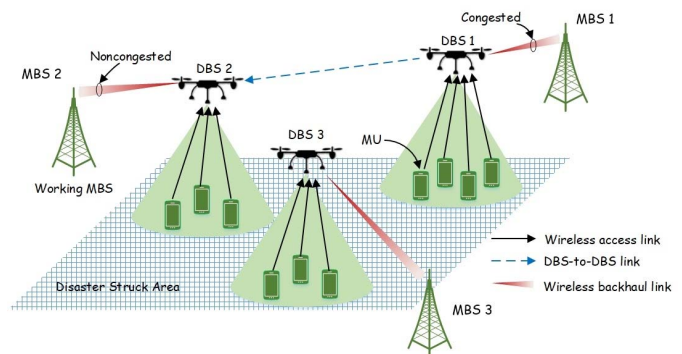


Fig. 1. Cooperative drones assisted mobile access network architecture

In order to relax the bottleneck in a wireless backhaul link, we propose a cooperative multiple drones assisted mobile access network architecture by deploying multiple DBSs in a disaster struck area. A DBS may relay data not only from its MUs, but also from its neighboring DBSs via DBS-to-DBS (D2D) communications. As shown in Fig. 1, each DBS communicates with its nearby working MBS to upload MUs' traffic. Assume that the wireless backhaul link between DBS 1 and working MBS 1 is the bottleneck, i.e., the capacity of the access links between DBS 1 and its MUs is larger than the capacity of the backhaul link between DBS 1 and its MBS. Then, DBS 1 can offload some traffic to DBS 2 via D2D communications. DBS 2 transmits the received traffic to its MBS via its backhaul link, which is not bottleneck. Here, we only consider the uplink scenario, i.e., MUs are uploading data streams to working MBSs via deployed DBSs.

The contributions of this paper are summarized as follows. 1) The cooperative drones assisted mobile access network architecture is proposed. 2) We formulate the DBSs configuration problem by jointly optimizing multiple DBSs placement, MU association, and bandwidth allocation to maximize the overall data rate of MUs in uploading data to their nearby working MBSs. 3) The COoperative DBS plAcement and CHannel allocation (COACH) algorithm is proposed to efficiently solve the problem, and the performance of COACH is validated via extensive simulations.

The rest of this paper is organized as follows. In Section II, we briefly review the related works. In Section III, we introduce the related models and formulate the DBS placement

This work was supported in part by the National Science Foundation under Award CNS-1814748 and OIA-1757207.

and channel allocation problem. In Section V, we design the COACH algorithm to solve the problem. In Section VI, we demonstrate the performance of COACH via simulations. The conclusion is presented in Section VII.

II. RELATED WORKS

Applying DBSs to forward data between MUs and MBSs has received much attention. Sun and Ansari [8] proposed a strategy of deploying a DBS in a hotspot area to enhance the data rates of MUs. In the strategy, the DBS placement and user association are jointly considered to maximize the total data rate from MUs. A 3D deployment strategy for multiple DBSs proposed by Zhang *et al.* [9] can meet the data rate requirements of the MUs in a hotspot area with the minimum number of DBSs. Applying drones to help MUs communicate with MBSs in disaster scenarios has been investigated. Rahman *et al.* [10] proposed a drone based ad-hoc network over a disaster struck area to maximize the throughput of drone based ad-hoc network by optimizing the placement of drones. To provision communications in large disaster areas, Narang *et al.* [11] proposed a buses-and-drones mobile edge infrastructure by placing DBSs and RRH-buses in different locations. Under the assumption that MUs tend to cluster in a disaster struck area, Hayajneh *et al.* [12] transformed a disaster scenario into a hotspot scenario. They evaluated the energy consumption of the deployed drones by varying the number and configurations of drones.

Although DBSs have been proposed to serve in disaster struck areas, current works do not consider D2D communications while DBSs are also serving MUs in the disaster struck area. In the cooperative multiple drones assisted mobile access network architecture, different DBSs are allowed to communicate with each other, and thus data from a congested DBS¹ can be transmitted to MBSs through other DBSs, which could increase the overall throughput.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In the cooperative multiple drones assisted mobile access network architecture, as shown in Fig. 1, there are a number of DBSs available to be deployed over a disaster struck area. Each DBS is to receive data from its associated MUs via wireless access links, receive/transmit data from/to other DBSs via D2D links, and transmit the received data to its associated MBS via the wireless backhaul link. Denote $\mathbf{I}$ and $\mathbf{J}$ as the set of DBSs and MUs in the disaster struck area, respectively, and i and j are used to index these DBSs and MUs, respectively.

A. Pathloss model

Each DBS may have three types of communications, i.e., MU-to-DBS, DBS-to-DBS, and DBS-to-MBS communications. In this section, we will explain the pathloss models applied in three types of communications, respectively.

¹A DBS is said to be congested if the total data rate from MUs to this DBS is larger than the data rate from this DBS to its nearby MBS.

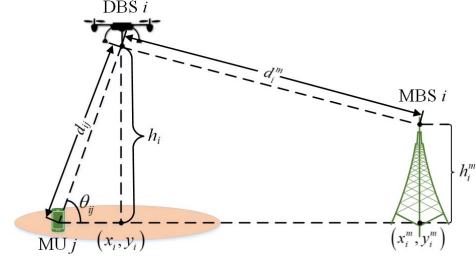


Fig. 2. Geometric relevance among a DBS, an MU, and an MBS

1) *MU-to-DBS pathloss model*: The link between an MU and its associated DBS is normally modeled as a probabilistic Line-of-Sight (LoS) link. Denote the pathloss of having LoS and Non-Line-of-Sight (NLoS) between DBS i and MU j (in dB) as η_{ij}^{LoS} and η_{ij}^{NLoS} , respectively, where [13]

$$\eta_{ij}^{LoS} = 20 \log \left(\frac{4\pi f d_{ij}}{c} \right) + \xi_{ij}^{LoS}, \quad (1)$$

$$\eta_{ij}^{NLoS} = 20 \log \left(\frac{4\pi f d_{ij}}{c} \right) + \xi_{ij}^{NLoS}. \quad (2)$$

Here, f is the carrier frequency, c is the speed of light, ξ_{ij}^{LoS} and ξ_{ij}^{NLoS} are the additional losses for LoS and NLoS, respectively, and d_{ij} is the distance between DBS i and MU j , i.e.,

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + h_i^2}, \quad (3)$$

where $\langle x_i, y_i, h_i \rangle$ is the location of DBS i and $\langle x_j, y_j \rangle$ is the location of MU j . The probability of having LoS between DBS i and MU j can be estimated by [13]

$$\rho_{ij} = \frac{1}{1 + b e^{-\beta(\theta_{ij} - b)}}, \quad (4)$$

where b and β are two environmental parameters in the disaster struck area, and θ_{ij} is the elevation angel between DBS i and MU j (which is indicated in Fig. 2), i.e., $\theta_{ij} = \arcsin \left(\frac{h_i}{d_{ij}} \right)$.

Based on Eq. (1), (2), and (4), the pathloss between DBS i and MU j , denoted as η_{ij}^c , can be modeled as [14]

$$\eta_{ij}^c = \rho_{ij} \eta_{ij}^{LoS} + (1 - \rho_{ij}) \eta_{ij}^{NLoS}. \quad (5)$$

Note that DBS i can communicate with MU j if the average pathloss between MU j and DBS i is less than a predefined threshold η^{th} (i.e., $\eta_{ij}^c < \eta^{th}$).

2) *DBS-to-MBS pathloss model*: Different from the channel between a DBS and an MU, the channel between a DBS and its associated working MBS is assumed to be free space [13]. Thus, the average pathloss between DBS i and its MBS is

$$\eta_i^m = 20 \log \left(\frac{4\pi f d_i^m}{c} \right), \quad (6)$$

where d_i^m is the distance between DBS i and its MBS, i.e.,

$$d_i^m = \sqrt{(x_i - x_i^m)^2 + (y_i - y_i^m)^2 + (h_i - h_i^m)^2} \quad (7)$$

where $\langle x_i^m, y_i^m, h_i^m \rangle$ is the 3-D position of DBS i 's associated working MBS.

3) *DBS-to-DBS pathloss model*: Links between DBSs are also considered as free space transmission, and thus the pathloss between DBS i and DBS i' (where $i' \in \mathbf{I}$) is

$$\eta_{ii'}^d = 20 \log \left(\frac{4\pi f_{ii'} d_{ii'}^d}{c} \right), \quad (8)$$

where $f_{ii'}$ is the carrier frequency used between DBS i and i' , and $d_{ii'}^d$ is the distance between DBS i and DBS i' , i.e., $d_{ii'}^d = \sqrt{(x_i - x_{i'})^2 + (y_i - y_{i'})^2 + (h_i - h_{i'})^2}$.

B. Data rate model

Each DBS is connected to a dedicated MBS, and each DBS is operated in the in-band full-duplex mode, i.e., the backhaul link (from a DBS to its MBS) and the access links (from MUs to their DBSs) can transmit data simultaneously over the same frequency band [15]. Note that different DBSs are signed different frequency bands but with the same bandwidth to enable their own access and backhaul communications.

For DBS $i \in \mathbf{I}$, its allocated frequency band is equally divided into a number of channels. Denote $\mathbf{K}_i$ as the set of these channels allocated to DBS i and k is used to index these channel. λ is the bandwidth of each channel. Denote u_{jk} as the binary variable to indicate whether channel k is allocated to MU j in uploading its data (i.e., $u_{jk} = 1$) or not (i.e., $u_{jk} = 0$). Denote v_{ik} as the binary variable to indicate whether channel k is allocated to DBS i in conducting backhaul communications or not. Denote $w_{ii'k}$ as the binary variable to indicate whether channel k is allocated to DBS i and DBS i' for D2D communications or not². Denote $\mathbf{K}$ as the set of all channels, i.e., $\mathbf{K} = \bigcup_{i \in \mathbf{I}} \mathbf{K}_i$.

1) *Data rate model between an MU and its DBS*: The data rate of MU j in uploading its data to DBS i is

$$r_{ij}^a = \sum_{k \in \mathbf{K}_i} (\lambda u_{jk} \log_2 (1 + \tau_{ijk})), \quad (9)$$

where τ_{ijk} is the SINR of DBS i in receiving signal from MU j over channel k , i.e.,

$$\tau_{ijk} = \frac{p_j g_{ij}}{\lambda N_0 + \gamma p_i}, \quad (10)$$

where p_j is the transmission power of MU j per channel, N_0 is the noise power, γ is the self-interference parameter, p_i is the transmission power of DBS i per channel, and g_{ij} is the channel gain³ between DBS i and MU j , i.e., $g_{ij} = 10^{-\frac{\eta_{ij}^a}{10}}$, where η_{ij}^a is the pathloss between DBS i and MU j .

2) *Data rate model between a DBS and its MBS*: The data rate from DBS i to its MBS can be expressed as

$$r_i^b = \sum_{k \in \mathbf{K}_i} (\lambda v_{ik} \log_2 (1 + \tau_i^b)), \quad (11)$$

where τ_i^b is the SNR for transmission from DBS i to its associated MBS over any channel, i.e., $\tau_i^b = \frac{p_i g_i^b}{\lambda N_0}$. Here, p_i is the transmission power of DBS i per channel, g_i^b is the channel gain between DBS i and its associated MBS, i.e.,

²If the channel $k \notin \mathbf{K}_i$, then $\forall i \in \mathbf{I}, j \in \mathbf{J}, v_{ik} = u_{jk} = 0$.

³For a clear exposition, the shadowing and fading effects are not considered in estimating the channel gain.

$g_i^b = 10^{-\frac{\eta_i^b}{10}}$, and η_i^b is the pathloss between DBS i and its associated MBS. Note that DBS i 's associated MBS may receive the interference from an MU (which is uploading data to DBS i over channel k) or a DBS (which is transmitting data to DBS i over channel k). Here, we do not consider the mentioned interference since the distance between DBS i and its associated MBS is much shorter than the distance between MUs/other DBSs and DBS i 's associated MBS.

3) *Data rate model of D2D communications*: D2D communications is conducted between DBS i and DBS i' . As mentioned before, two DBSs are operated in two different spectrum bands to conduct their access and backhaul communications. Sender DBSs could reuse spectrum bands of receiver DBSs to achieve D2D communications. That is, any channel $k \in \mathbf{K}_{i'}$ can be used to conduct the communication from DBS i to DBS i' . Thus, the data rate of transmitting data from DBS i to DBS i' is

$$r_{ii'}^d = \sum_{k \in \mathbf{K}_{i'}} (\lambda w_{ii'k} \log_2 (1 + \tau_{ii'k}^d)), \quad (12)$$

where $\tau_{ii'k}^d$ is the SINR between DBS i and DBS i' over channel k ($k \in \mathbf{K}_{i'}$), i.e.,

$$\tau_{ii'k}^d = \frac{p_i g_{ii'}^d}{\lambda N_0 + \gamma p_{i'} v_{i'k}}, \quad (13)$$

where $g_{ii'}^d$ is the channel gain between DBS i and DBS i' , i.e., $g_{ii'}^d = 10^{-\frac{\eta_{ii'}^d}{10}}$ (where $\eta_{ii'}^d$ is the pathloss between two DBSs, which is obtained from Eq. (8)). Here, $\gamma p_{i'} v_{i'k}$ is the self-interference from DBS i' in conducting backhaul communications if it uses the same channel k , where $p_{i'}$ is the transmission power of DBS i' and $v_{i'k}$ is the indication of channel k allocated to DBS i' for its backhaul. Here, if channel k is used for MU j in uploading its data to DBS i' , channel k cannot be used for the D2D communications, i.e.,

$$u_{jk} w_{ii'k} = 0, k \in \mathbf{K}_{i'}. \quad (14)$$

C. Problem formulation

We formulate the DBS placement and channel allocation problem as follows.

$$\mathbf{P0}: \arg \max_{u_{jk}, v_{ik}, w_{ii'k}, x_i, y_i} \sum_{i \in \mathbf{I}} \sum_{j \in \mathbf{J}} r_{ij}^a \quad (15)$$

$$s.t. : C1 : \sum_{j \in \mathbf{J}} r_{ij}^a + \sum_{i' \in \mathbf{I} \setminus i} r_{ii'}^d \leq r_i^b + \sum_{i' \in \mathbf{I} \setminus i} r_{ii'}^d, \forall i \in \mathbf{I},$$

$$C2 : u_{jk} w_{ii'k} = 0, \forall k \in \mathbf{K}_{i'}, \forall i, i' \in \mathbf{I}, \forall j \in \mathbf{J},$$

$$C3 : \sum_{i \in \mathbf{I}} r_{ij}^a \geq r_j', \forall j \in \mathbf{J}, \exists u_{jk} \neq 0,$$

$$C4 : u_{jk}, v_{ik}, w_{ii'k} \in \{0, 1\}, \forall k \in \mathbf{K}_{i'}, \forall i, i' \in \mathbf{I}, \forall j \in \mathbf{J},$$

where the objective is to maximize the throughput of the network, C1 imposes the sum of the incoming data rate to be less than the sum of the outgoing data rate for each DBS, i.e., the backhaul link is not the bottleneck link. C2 imposes that Eq. (14) is satisfied. C3 imposes that the provisioned data rate of MU j to be larger than its data rate requirement, which is denoted as r_j' . C4 imposes that u_{jk} , v_{ik} , and $w_{ii'k}$ to be binary variables.

IV. COOPERATIVE DBS PLACEMENT AND CHANNEL ALLOCATION

We design a heuristic algorithm, i.e., COoperative DBS plAcement and CHannel allocation (COACH), to efficiently solve **P0**. The basic idea of COACH is to decompose **P0** into two sub-problems, i.e., DBS placement and channel allocation.

Algorithm 1 COACH

```

1: for all DBSs  $i \in \mathbf{I}$  do
2:   for all locations  $n \in \mathbf{N}$  do
3:     Obtain  $\mathbf{J}_i^n$ .
4:     Calculate  $\bar{r}_i^a$  and  $\bar{r}_i^b$  based on Eq. (17) and Eq. (18).
5:     Calculate  $\bar{r}_i^n$  based Eq. (19)
6:   end for
7:   Select the best location  $n^*$  for DBS  $i$  based on Eq. (20).
8:    $\forall j \in \mathbf{J}_i^n$ , calculate  $z_{ij}$  based on Eq. (21).
9:   Allocate channels to MUs iteratively based on  $g_{ij}$ .
   Channel allocation terminates once  $\sum_{j \in \mathbf{J}_i^n} z_{ij} \geq |\mathbf{K}_i|$  or
   all MUs are assigned channels.
10:  Remove the MUs (which are assigned channels) from  $\mathbf{J}_i^n$ .
11:  Calculate  $r_i^a$  and  $c_i^b$  based on Eq. (22) and Eq. (23).
12: end for
13: Divide DBSs into two sets, i.e.,  $\mathbf{I}^s$  and  $\mathbf{I}^r$ , where  $\mathbf{I}^s = \{i \in \mathbf{I} | r_i^a > c_i^b\}$  and  $\mathbf{I}^r = \{i \in \mathbf{I} | r_i^a < c_i^b\}$ .
14: Derive the number of available channels for each DBS in  $\mathbf{I}^r$  based on Eq. (24).
15: Calculate the data rate of all the possible D2D pairs based on Eq. (25).
16: Calculate the number of channels could be allocated for all the possible D2D pairs based on Eq. (26).
17: while  $\mathbf{I}^r \neq \emptyset$  and  $\mathbf{I}^s \neq \emptyset$  do
18:   Select the D2D pair that incurs the maximum data rate and allocate the number of channels.
19:   Adjust the MU association of the source DBS for the selected D2D pair such that Constraint C1 is met.
20:   Remove the source and destination DBSs of the selected D2D pair from  $\mathbf{I}^s$  and  $\mathbf{I}^r$ , respectively.
21: end while

```

A. DBS placement

The disaster struck area is divided into a number of locations with the same size, denoted by set $\mathbf{N}$. If DBS i is placed over location n (where $n \in \mathbf{N}$), then the 2-D location of DBS i (i.e., $\langle x_i, y_i \rangle$) is the center of location n . The DBS placement is to find the location that yields the maximum estimated data rate for MUs to upload their data streams. Specifically,

- 1) Denote $\mathbf{J}_i^n$ as the set of MUs satisfying the channel gain threshold⁴ when DBS i is at location n , (i.e., $g_{ij}^n > g_i^{th}$). Assume that all MUs in $\mathbf{J}_i^n$ are associated to DBS i . The average channel gain between the MUs in $\mathbf{J}_i^n$ and DBS i is defined as

⁴Channel gain threshold defines as the minimum channel gain between a DBS and an MU to enable communications.

$$\bar{g}_i^n = \sum_{j \in \mathbf{J}_i^n} g_{ij}^n / |\mathbf{J}_i^n|, \quad (16)$$

Thus, the estimated access link data rate is defined as

$$\bar{r}_i^a = |\mathbf{K}_i| \lambda \log_2 \left(1 + \frac{p_j \bar{g}_i^n}{|\mathbf{K}_i| \lambda N_0} \right). \quad (17)$$

The data rate of backhaul link between DBS i and its MBS is defined as

$$\bar{r}_i^b = |\mathbf{K}_i| \lambda \log_2 \left(1 + \frac{p_i \bar{g}_i^m}{|\mathbf{K}_i| \lambda N_0} \right), \quad (18)$$

where g_i^m is the channel gain between DBS i and its associated MBS. Thus, the estimated data rate of the MUs (in $\mathbf{J}_i^n$) in uploading their data to the MBS via DBS i , which is deployed over location n ($\bar{r}_i^n$), is

$$\bar{r}_i^n = \min \{ \bar{r}_i^a, \bar{r}_i^b \}, \quad (19)$$

- 2) We iteratively deploy DBS i over each location and find the location that incurs the maximum value of $\bar{r}_i^n$, i.e.,

$$n^* = \arg \max \{ \bar{r}_i^n | n \in \mathbf{N} \}. \quad (20)$$

DBS i will be placed over location n^* .

B. Channel allocation

1) *Access link channel allocation:* After placing DBS i , the associated MUs will be allocated channels. Note that MUs that cannot be served⁵ will be removed from $\mathbf{J}_i^n$. Specifically,

- (a) Calculate the minimum number of channels for MU j in uploading data to DBS i (denoted as z_{ij}) in order to satisfy the data rate requirement of MU j (i.e., C3) based on Eq. (9), i.e.,

$$z_{ij} = \left\lceil \frac{r_j'}{\lambda \log_2 (1 + \tau_{ijk})} \right\rceil. \quad (21)$$

- (b) The available channels may not be enough to satisfy all the MUs' requirements. Thus, the MU, which incurs the highest channel gain to DBS i , will be assigned channels first. The access link channel allocation will be terminated until all the MUs in $\mathbf{J}_i^n$ meet their requirements or there is no available channel.
- (c) Update the set of $\mathbf{J}_i^n$ by removing the MUs that are not assigned with channels in Step (b).

The data rate of DBS i in receiving data from MUs (r_i^a) is,

$$r_i^a = \sum_{j \in \mathbf{J}_i^n} r_{ij}^a, \quad (22)$$

where r_{ij}^a is the data rate of MU j in uploading data to DBS i . The capacity of the backhaul link between DBS i and its MBS is

$$c_i^b = |\mathbf{K}_i| \lambda \log_2 (1 + \tau_{ik}^m). \quad (23)$$

Thus, DBS i is categorized to three kinds:

- $r_i^a < c_i^b$: DBS i is not using all backhaul channels. Thus, DBS i may help other DBSs offload their traffic by using D2D communications (i.e., **DBS i may be the receiver of the D2D communications**).
- $r_i^a = c_i^b$: DBS i 's access link data rate just matches its backhaul link capacity. That is, DBS i does not have to offload its traffic to other DBSs; also, it cannot accept traffic offloaded from other DBSs.

⁵An MU unable to be served means that the wireless channels allocated to the MU are not enough to satisfy the data rate requirement of the MU.

- $r_i^a > c_i^b$: the backhaul link of DBS i is the bottleneck, and so it has to offload traffic to other DBSs via D2D communications. That is, **DBS i could be the transmitter of D2D communications.**

2) *DBS to DBS communications channel allocation*: After placing all DBSs and allocating access channels, we allocate channels for D2D communications. We divide the placed DBSs into three sets, i.e., the set of DBSs with bottleneck backhaul links, i.e., $\mathbf{I}^s = \{i \in \mathbf{I} \mid r_i^a > c_i^b\}$, the set of DBSs with their backhaul links are not congested, i.e. $\mathbf{I}^r = \{i \in \mathbf{I} \mid r_i^a < c_i^b\}$, and the set of DBSs with their access link data rates just matched with their respective backhaul link capacities, i.e., $\mathbf{I}^o = \{i \in \mathbf{I} \mid r_i^a = c_i^b\}$. D2D channel allocation is to select a DBS in $\mathbf{I}^s$ as the source node and a DBS in $\mathbf{I}^r$ as the destination node, and allocate channels to offload traffic from the source node to the destination node. Specifically,

- (a) A D2D communications pair (from DBS i to DBS i') uses the unused channels of DBS i' , i.e., if there are some unused channels in the destination node's access links, these channels can be used for D2D communications. Thus, we derive the number of unused channels (in the access links) for each DBS in $\mathbf{I}^r$ as

$$z_{i'}^r = |\mathbf{K}_{i'}| - \sum_{j \in \mathbf{J}_{i'}^n} z_{i'j}. \quad (24)$$

- (b) Let $|\mathbf{I}^s|$ and $|\mathbf{I}^r|$ be the number of DBSs in $\mathbf{I}^s$ and $\mathbf{I}^r$, respectively. Thus, there are $|\mathbf{I}^s| \times |\mathbf{I}^r|$ number of possible D2D communications. For each possible D2D communications (from DBS i to DBS i'), we derive the average SINR as $\tau_{ii'}^d = \sum_{k \in \mathbf{K}_{i'}} \frac{\tau_{ii'k}^d}{|\mathbf{K}_{i'}|}$, where $\tau_{ii'k}^d$ is calculated based on Eq. (13). Thus, the data rate of D2D pair (DBS i and i') is

$$r_{ii'}^d = \lambda z_{ii'}^d \log_2(1 + \tau_{ii'}^d), \quad (25)$$

where $z_{ii'}^d$ is the number of channels used for D2D communications from DBS i to i' , which depends on the number of unused channels in DBS i' 's access link (i.e., $z_{i'}^r$ in Eq. (24)) as well as the amount of traffic to be offloaded from DBS i (i.e., $r_i^a - c_i^b$). Thus, we have

$$z_{ii'}^d = \min \left\{ z_{i'}^r, \left\lceil \frac{r_i^a - c_i^b}{\lambda \log_2(1 + \tau_{ii'}^d)} \right\rceil \right\}, \quad (26)$$

where $\left\lceil \frac{r_i^a - c_i^b}{\lambda \log_2(1 + \tau_{ii'}^d)} \right\rceil$ is the number of required channels to offload $r_i^a - c_i^b$ amount of traffic from DBS i .

- (c) We calculate the data rates of all possible D2D pairs based on Eq. (25), pick the D2D pair, which provisions the maximum data rate, allocate the number of channels to the D2D pair (based on Eq. (26)), and remove the source DBS and destination DBS in the D2D pair from $\mathbf{I}^s$ and $\mathbf{I}^r$, respectively. Note that, even if we enable D2D communications to offload the traffic from source DBS i to destination DBS i' , source DBS i still may not satisfy Constraint C1. If so, we will iteratively un-associate MUs (which are associated with source DBS i) to reduce access link traffic (i.e., the value of r_i^a) until $r_i^a \leq c_i^b + r_{ii'}^d$. The MU with a lower value of r_{ij}^a will be disassociated from source DBS i first.

TABLE I
SIMULATION PARAMETERS

Disaster area	2 km $\times$ 2 km
Location area	100 m $\times$ 100 m
Carrier frequency (f)	2 GHz
Environment index (b)	9.61
Environment index (β)	0.16
ξ^{LoS}	1 dB [13]
ξ^{NLoS}	20 dB [13]
γ	130 dB [16]
N_0	-104 dBm/Hz
h_i^m	30 m
h_i	50 m

V. SIMULATION RESULTS

Assume that 6 DBSs will be deployed over a disaster struck area. The altitude of DBSs are the same. There are 25 channels available for each DBS and the bandwidth of each channel is 180 kHz. The data rate requirements of MUs are generated based on a normal distribution, i.e., $N(2, 1)$ Mbps. Other simulation parameters are listed in Table I.

The results of COACH are compared to two other reference algorithms, i.e., access link aware (ALA) and backhaul link aware (BLA) DBS placement and channel allocation. ALA is to maximize the access link data rate of a DBS without considering the backhaul link capacity. BLA is to maximize the overall throughput by jointly considering the access link data rate and the backhaul link capacity. Both algorithms do not apply D2D communications to offload traffic.

Fig. 3 shows the total data rate (of uploading data from MUs to MBSs) by varying the number of MUs. Note that the number of MUs in each location is selected based on a normal distribution, i.e., $N(\psi, 5)$, where ψ is the average number of MUs in each location. Thus, we vary ψ to adjust the number of MUs. From the figure, we can see that COACH always yields the highest total data rate. However, as the number of MUs increases, the gap of the total data rate between COACH and BLA reduces because as the number of MUs is small, the backhaul link utilization incurred by BLA/ALA is not balanced among DBSs (e.g., some DBSs have 100% backhaul link utilization while some have the backhaul link utilization less than 80%); yet, COACH balances the backhaul link utilization by offloading the traffic from a DBS with congested backhaul link to a DBS with non-congested backhaul link, thus increasing the total data rate. ALA is not affected much by the user density because ALA only maximizes the access link data rate without considering the backhaul link, and thus backhaul links have high probability of becoming bottleneck. However, as the number of MUs increases, backhaul link utilization incurred by both ALA and BLA will be more balanced, as demonstrated in Fig. 4. That is, the backhaul link, whose link utilization was less than 100%, is approaching 100%, thus reducing the average backhaul link utilization gap among COACH, ALA and BLA.

Fig. 5 shows the performance of different algorithms by varying the altitude of DBSs. Note that the altitude of MBSs is 30 m, and thus decreasing the altitude of a DBS from 50 m to 30 m essentially reduces the distance between a DBS

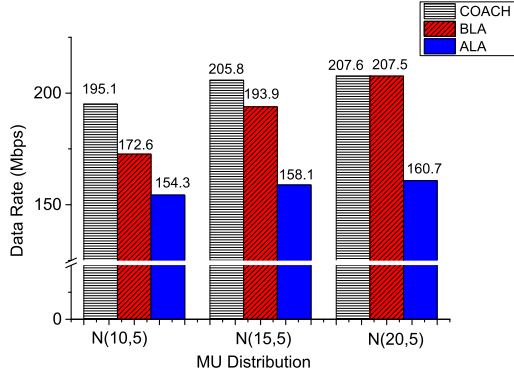


Fig. 3. Total data rate over different MU distributions.

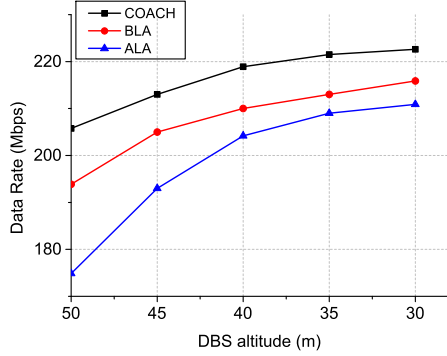


Fig. 5. Total data rate over different DBS altitudes.

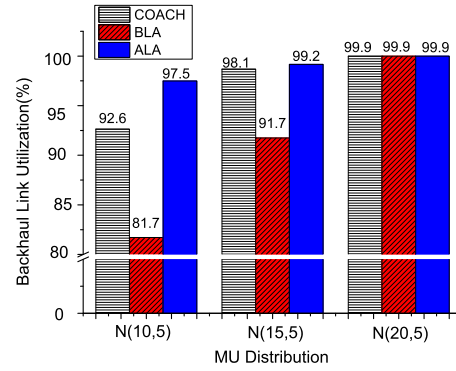


Fig. 4. Backhaul link utilization over different MU distributions.

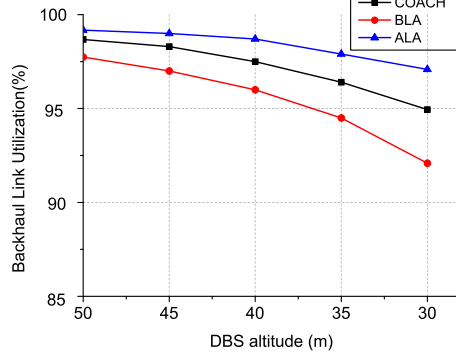


Fig. 6. Backhaul link utilization over different DBS altitudes.

and its MBS. Hence, the backhaul link between a DBS and its MBS has better channel gain, thus incurring larger backhaul link capacity. In Fig. 5, COACH incurs the highest data rate because it offloads traffic from congested backhaul links to non-congested backhaul links via D2D communications. Increasing the backhaul link capacity enables COACH to offload more traffic, thus further increasing the overall throughput. BLA and ALA, on the other hand, do not offload traffic among backhaul links, and thus they incur lower data rates. Fig. 6 shows how the altitude of the DBS affects the average backhaul link utilization. We can see that the average backhaul link utilization incurred by the three algorithms reduces as the altitude of the DBS reduces because decreasing the altitude of a DBS increases the capacity of the backhaul link, thus reducing the backhaul link utilization.

VI. CONCLUSIONS

In this paper, we have formulated the joint DBS placement and channel allocation problem and designed the COACH algorithm to efficiently solve the problem. The performance of COACH has been demonstrated via extensive simulations.

REFERENCES

- [1] A. Fotouhi, M. Ding, and M. Hassan, "Flying drone base stations for macro hotspots," *IEEE Access*, vol. 6, pp. 19 530–19 539, 2018.
- [2] N. Ansari and X. Sun, "Mobile edge computing empowers internet of things," *IEICE Trans. Comm.*, vol. 101, no. 3, pp. 604–619, 2018.
- [3] N. Zhao, W. Lu, M. Sheng, Y. Chen, J. Tang, F. R. Yu, and K. Wong, "Uav-assisted emergency networks in disasters," *IEEE Wireless Communications*, vol. 26, no. 1, pp. 45–51, February 2019.
- [4] S. Liu, Z. Wei, Z. Guo, X. Yuan, and Z. Feng, "Performance analysis of UAVs assisted data collection in wireless sensor network," in *IEEE Vehicular Technology Conf. (VTC Spring)*, June 2018, pp. 1–5.
- [5] L. Zhang and N. Ansari, "A framework for 5G networks with in-band full-duplex enabled drone-mounted base-stations," *IEEE Wireless Commun.*, pp. 1–7, 2019.
- [6] W. Shi, J. Li, W. Xu, H. Zhou, N. Zhang, S. Zhang, and X. Shen, "Multiple drone-cell deployment analyses and optimization in drone assisted radio access networks," *IEEE Access*, vol. 6, pp. 12 518–12 529, 2018.
- [7] M. Mozaffari, W. Saad, M. Bennis, Y. Nam, and M. Debbah, "A tutorial on uavs for wireless networks: Applications, challenges, and open problems," *IEEE Communications Surveys Tutorials*, pp. 1–1, 2019.
- [8] X. Sun and N. Ansari, "Jointly optimizing drone-mounted base station placement and user association in heterogeneous networks," in *2018 IEEE International Conf. on Comm. (ICC)*, May 2018, pp. 1–6.
- [9] S. Zhang, X. Sun, and N. Ansari, "Placing multiple drone base stations in hotspots," in *2018 IEEE 39th Sarnoff Symposium*, Sept 2018, in press.
- [10] S. Rahman, G. Kim, Y. Cho, and A. Khan, "Positioning of UAVs for throughput maximization in software-defined disaster area UAV communication networks," *J. Comm. and Netw.*, vol. 20, no. 5, pp. 452–463, Oct 2018.
- [11] M. Narang, W. Liu, J. Gutierrez, and L. Chiaraviglio, "A cyber physical buses-and-drones mobile edge infrastructure for large scale disaster emergency communications," in *2017 IEEE 37th Int. Conf. Distributed Comput. Systems Workshops (ICDCSW)*, June 2017, pp. 53–60.
- [12] A. M. Hayajneh, S. A. R. Zaidi, D. C. McLernon, M. D. Renzo, and M. Ghogho, "Performance analysis of UAV enabled disaster recovery networks: A stochastic geometric framework based on cluster processes," *IEEE Access*, vol. 6, pp. 26 215–26 230, 2018.
- [13] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Comm. Lett.*, vol. 3, no. 6, pp. 569–572, Dec 2014.
- [14] E. Kalantari, H. Yanikomeroglu, and A. Yongacoglu, "On the number and 3D placement of drone base stations in wireless cellular networks," in *IEEE Vehicular Technology Conf. (VTC-Fall)*, Sep 2016, pp. 1–6.
- [15] R. Rajashekar, M. D. Renzo, K. V. S. Hari, and L. Hanzo, "A generalized transmit and receive diversity condition for feedback-assisted MIMO systems: Theory and applications in full-duplex spatial modulation," *IEEE Trans. Signal Process.*, vol. 65, no. 24, pp. 6505–6519, Dec 2017.
- [16] Y. Choi and H. Shirani-Mehr, "Simultaneous transmission and reception: Algorithm, design and system level performance," *IEEE Trans. on Wireless Comm.*, vol. 12, no. 12, pp. 5992–6010, December 2013.